

MODULE 2 SYLLABUS

Azure Storage

Storage Basics – Storage Types – Azure Storage Offerings – Understanding Azure Regions – Using Storage Accounts – Enabling Larger and Faster Storage – Resizing Azure Disks – Using Premium Storage – Monitoring Azure Storage Accounts – Best Practices for Azure Storage – Azure VM Storage Types – Azure Files – Managing Azure Storage.

Day 1: Storage Account Fundamentals (50 minutes)

Task 1: Create Multiple Storage Account Types

Create three different storage accounts to understand their differences:

Storage Account 1: General Purpose v2

- Name: gp2storage[yourname][randomnumber]
- Resource Group: Storage-Lab-RG
- Region: Your nearest region
- Performance: Standard
- Redundancy: Locally-redundant storage (LRS)
- Access tier: Hot

Storage Account 2: Premium Block Blobs

- Name: premblob[yourname][randomnumber]
- Performance: Premium
- Premium account type: Block blobs
- Redundancy: LRS

Storage Account 3: General Purpose v2 with GRS

- Name: grstorage[yourname][randomnumber]
- Performance: Standard
- Redundancy: Geo-redundant storage (GRS)
- Access tier: Cool

Task 2: Understanding Storage Types

Document the following for each storage account:

1. What services are available (Blob, File, Queue, Table)?
2. Pricing differences
3. Performance characteristics
4. Appropriate use cases

Questions to Answer:

- When would you choose Standard vs Premium performance?
 - What's the difference between Hot, Cool, and Archive access tiers?
 - Explain LRS, ZRS, GRS, and GZRS redundancy options
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Day 2: Azure Regions and Data Replication (40 minutes)

Task 1: Explore Region Pairs

1. Navigate to your GRS storage account
2. Find the primary and secondary regions
3. Document the region pair for your chosen region

Task 2: Test Geo-Replication

1. Upload 5 sample files (images, documents) to a blob container in your GRS storage account
2. Container name: "testdata"
3. Check the replication status
4. **Question:** How long does it take for data to replicate to the secondary region?

Task 3: Region Selection Exercise

Research and document:

1. List 3 Azure regions closest to your location
2. Compare pricing for storage in these regions (use Azure Pricing Calculator)
3. Check service availability - are Premium Storage and availability zones available?
4. Recommend the best region for:
 - A low-cost backup solution
 - A high-performance database application
 - A globally distributed web application

Day 3: Working with Storage Accounts (60 minutes)

Task 1: Upload and Manage Blobs

1. In your GP2 storage account, create containers:
 - Container 1: "documents" (Private access)
 - Container 2: "public-images" (Blob public read access)
 - Container 3: "archive-data" (Private access)
2. Upload different file types:
 - 3 images to public-images
 - 5 documents (PDF, Word, Excel) to documents
 - 2 large files (>100MB) to documents
3. Generate a SAS token for the "documents" container:
 - Valid for 2 hours
 - Read and List permissions only
 - Test the SAS URL in a browser

Task 2: Access Tiers and Lifecycle Management

1. Change the access tier of one blob from Hot to Cool
2. Set up a lifecycle management rule:
 - a. Move blobs in "archive-data" to Cool tier after 30 days
 - b. Move to Archive tier after 90 days
 - c. Delete after 365 days
3. **Document:** Screenshot of the lifecycle management policy

Task 3: Storage Explorer

1. Download and install Azure Storage Explorer
2. Connect to your storage account
3. Perform the following operations:
 - Upload a folder with multiple files
 - Download a blob
 - Copy blobs between containers
 - Generate and test a SAS token

Screenshot Required: Storage Explorer interface showing your connected storage accounts

Day 4: Azure VM Storage and Disk Management (70 minutes)

Task 1: Create VM with Custom Disk Configuration

1. Create a new VM (either Windows or Linux):
 - Name: StorageTestVM

- Size: Standard_B2s
 - OS Disk: Premium SSD, 128 GB
 - Enable "Delete with VM" for OS disk
2. Document the disk performance metrics:
 - IOPS limit
 - Throughput limit
 - Disk type and pricing

Task 2: Add and Configure Data Disks

Add three data disks with different configurations:

Disk 1: Standard HDD

- Size: 512 GB
- Name: StandardHDD-Disk
- Host caching: None

Disk 2: Standard SSD

- Size: 256 GB
- Name: StandardSSD-Disk
- Host caching: Read-only

Disk 3: Premium SSD

- Size: 128 GB
 - Name: PremiumSSD-Disk
 - Host caching: Read/write
3. Initialize and format all disks inside the VM
 4. Create a comparison table showing IOPS and throughput for each

Task 3: Resize Azure Disks

1. Stop (deallocate) your VM
2. Resize the StandardSSD-Disk from 256 GB to 512 GB
3. Document the process and any limitations
4. Start the VM and extend the volume inside the OS

Questions:

- Can you shrink a disk in Azure?
- What happens to data when you resize a disk?
- Can you resize a disk while the VM is running?

Task 4: Enable Ultra Disk (Documentation Exercise)

Research and document:

1. What are Ultra Disks and their use cases?
2. Performance characteristics (IOPS, throughput)
3. Region and VM size requirements
4. Cost comparison with Premium SSD

Day 5: Premium Storage Deep Dive (45 minutes)

Task 1: Premium Storage Performance Testing

1. Create a Premium Storage account
2. Upload a 500MB file and measure the upload time
3. Download the same file and measure the download time
4. Compare with Standard Storage upload/download times

Task 2: Premium Files Share

1. Create a Premium Files share:
 - Name: premiumshare
 - Quota: 100 GB
 - Protocol: SMB
2. Mount the file share on your VM:
 - For Windows: Map as network drive
 - For Linux: Mount using CIFS
3. Copy 10 files to the Premium share
4. Test read/write performance

Task 3: Premium Storage Use Cases

Document three real-world scenarios where Premium Storage is essential:

- Include workload type
 - Performance requirements
 - Cost justification
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Day 6: Azure Files Implementation (50 minutes)

Task 1: Create Standard Azure Files Share

1. In your GP2 storage account, create a file share:

- Name: companyshare
 - Quota: 50 GB
 - Tier: Transaction optimized
2. Create the following folder structure:

```
/departments
  /hr
  /finance
  /it
/shared
  /templates
  /policies
```

3. Upload sample files to each folder

Task 2: Mount Azure Files Share

Mount the file share on:

1. Windows VM (persistent mapping that survives reboots)
2. Document the PowerShell/CMD commands used
3. Test access by creating and editing files

Task 3: Azure Files Snapshots

1. Create a share-level snapshot
2. Modify or delete some files in the share
3. Restore files from the snapshot
4. **Document:** Screenshot showing snapshot restoration

Task 4: Azure File Sync (Research)

Research and document:

1. What is Azure File Sync?
2. Architecture components (sync agent, storage sync service)
3. Use case: Multi-site file synchronization scenario
4. Cloud tiering benefits.

Day 7: Monitoring Azure Storage (45 minutes)

Task 1: Enable Storage Analytics

1. Enable diagnostics logging for your GP2 storage account:
 - Blob service logs
 - Retention: 7 days
2. Enable Storage Analytics metrics:
 - Transaction metrics
 - Capacity metrics

Task 2: Create Monitoring Dashboard

1. Navigate to Metrics in Azure Monitor
2. Create charts for:
 - Total ingress/egress (last 24 hours)
 - Transaction count by API name
 - Success rate percentage
 - Average E2E latency
3. Pin these charts to a custom dashboard named "Storage Monitoring"

Screenshot Required: Complete monitoring dashboard

Task 3: Set Up Alerts

Create alerts for:

1. Storage account availability drops below 99%
2. Total egress exceeds 100 GB in 1 hour
3. Failed transactions exceed 50 in 15 minutes

Document the alert configuration and action group setup.

Task 4: Review Storage Insights

1. Navigate to Storage Insights (if available)
2. Explore:
 - Performance metrics
 - Availability metrics
 - Capacity trends
3. Generate a weekly report summary

Day 8: Best Practices Implementation (40 minutes)

Task 1: Security Best Practices

Implement the following security measures on your GP2 storage account:

1. **Disable public access** to blobs (account level)
2. **Enable secure transfer required** (HTTPS only)
3. **Configure network rules:**
 - Add your current IP to allowed IPs
 - Test access from allowed and blocked IPs
4. **Enable Azure Defender** for Storage (if budget allows)
5. **Configure access keys:**
 - Regenerate one access key
 - Test that applications can still connect using the other key

Task 2: Cost Optimization

1. Identify and document:
 - Storage accounts with no activity in 30 days
 - Blobs that can be moved to Cool tier
 - Orphaned disks (not attached to any VM)
2. Calculate potential savings by:
 - Moving 100 GB from Hot to Cool tier
 - Moving 500 GB from Cool to Archive tier
 - Deleting 5 orphaned managed disks (128 GB each)

Task 3: Backup and DR Strategy

Design and document:

1. Backup strategy for Azure Files share
2. Disaster recovery plan for critical storage accounts
3. RPO (Recovery Point Objective) and RTO (Recovery Time Objective) targets
4. Testing schedule for DR procedures

Task 4: Performance Optimization

Document best practices for:

1. Choosing appropriate storage redundancy
 2. Selecting the right disk type for different workloads
 3. Optimizing network performance for storage access
 4. Using CDN for frequently accessed blobs
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Day 9: Cleanup and Cost Analysis (30 minutes)

Task 1: Cost Review

1. Navigate to Cost Management
2. Filter by Storage-Lab-RG resource group
3. Create a cost breakdown showing:
 - Storage account costs
 - Disk costs
 - Data transfer costs
 - Total daily cost
4. Export cost data to Excel and create a pie chart

Task 2: Resource Inventory

Create a comprehensive inventory document listing:

- All storage accounts created (name, type, region, redundancy)
- All VMs and their attached disks
- All file shares
- All containers and approximate data stored

Task 3: Proper Cleanup

Delete resources in this order:

1. Delete all VMs (and associated resources)
2. Delete orphaned disks
3. Delete file shares
4. Delete blob containers
5. Delete storage accounts
6. Delete the resource group

Verify all resources are removed to avoid unexpected charges.

Deliverables

Submit a comprehensive report (PDF format) containing:

1. Technical Documentation (60%)

- Screenshots of all major tasks (minimum 25 screenshots)

- Configuration details for each storage account created
- Comparison tables for storage types and performance
- Monitoring dashboard and alerts setup
- Network architecture diagrams for storage access

2. Written Analysis (25%)

- Answers to all questions throughout the assignment
- Storage type comparison matrix (when to use what)
- Best practices summary (500 words)
- Cost optimization recommendations
- Security implementation checklist

3. Case Study (10%)

Write a 600-800 word case study proposing a complete storage solution for:

Scenario: A healthcare company needs to:

- Store 10 TB of patient records (rarely accessed, 7-year retention)
- Host a file share for 200 employees across 5 offices
- Run a high-performance database application
- Backup 50 VMs daily
- Ensure HIPAA compliance and disaster recovery

Your solution should include:

- Storage account types and configurations
- Redundancy choices with justification
- Cost estimates (monthly)
- Security measures
- Backup and DR strategy

4. Cost Analysis Report (5%)

- Detailed cost breakdown of all resources used
- Monthly cost projection for 24/7 operation
- Cost optimization recommendations
- ROI analysis for Premium vs Standard storage

Resources and References

- Azure Storage Documentation: <https://docs.microsoft.com/azure/storage/>
- Azure Storage Explorer: Download from Microsoft

- Azure Pricing Calculator: <https://azure.microsoft.com/pricing/calculator/>
 - Storage performance best practices documentation
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Submission Checklist

- All screenshots properly labeled and organized
- All questions answered completely
- Comparison tables and charts included
- Monitoring dashboard screenshot
- Cost analysis spreadsheet
- Case study completed
- All resources cleaned up (screenshot of empty resource group)
- Document formatted professionally with table of contents
- File named: AzureStorage_Assignment_[YourName]_[Date].pdf

Good luck! This comprehensive assignment will give you production-ready Azure Storage skills.